Problem 2 — The Paper Plane Challenge

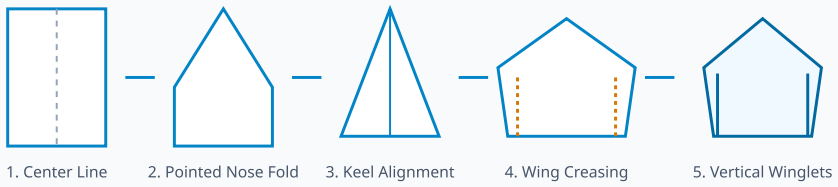
AERODYNAMIC OPTIMIZATION FOR MAXIMUM FLIGHT DURATION (HANG TIME)

ASSESSMENT	TOTAL MARKS	AIRCRAFT CLASS	PRIMARY METRIC
Engineering Challenge	25 (+5 Bonus)	Unite Glider	Time Aloft (Hang Time)

PART A — DESIGN & CONSTRUCTION

1. Construction Sequence & Structural Geometry

The aircraft ("Unite Plane") is constructed from a single standard A4 sheet (80 g/m², 210 mm × 297 mm) without external tape, adhesives, or fasteners. The design features a sharply pointed nose tapering into wide, broad main wings with folded vertical winglets at both outer tips, supported by a central structural keel.



2. Aerodynamic Choice & Configuration Rationale

To maximize time aloft, the design favors aerodynamic glide efficiency over raw velocity. The geometric principles behind the Unite plane include:

- **Expanded Planform Area & Low Wing Loading:** Spreading the aircraft mass across a wide wing area minimizes wing loading ($W_L = m / S$), reducing the minimum stall speed required to remain airborne.
- **Forward Swept Pointed Nose:** Concentrates layered paper folds toward the front tip, placing the Center of Gravity (CG) forward of the Center of Pressure (CP) for longitudinal stability.
- **Vertical Winglets (Tip Stabilizers):** Vertical folds at the outer wing edges suppress wingtip vortices, reducing induced drag during slow glides.
- **Dihedral Angle & Elevon Up-Trim:** An upward wing V-angle provides passive lateral (roll) stability, while a minor upward bend on the trailing edge creates restoring nose-up pitch torque.

PART B — AERODYNAMICS & FORCE ANALYSIS

High Area	Minimized	Moderate
WING SURFACE (S)	WING LOADING (W_L)	ASPECT RATIO (AR)

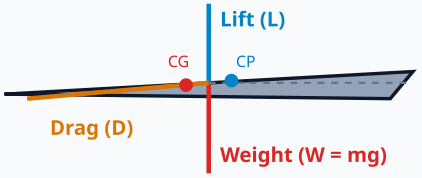
1. Mechanics of Lift Generation

Lift (L) is generated via combined static pressure differential across the wing profile (Bernoulli's Principle) and downward momentum deflection of oncoming air at an angle of attack α (Newton's Third Law):

$$L = 0.5 \times \rho \times v^2 \times S \times C_L \quad | \quad W_L = m / S$$

2. Four Forces Across Flight Phases

During the **Launch Phase**, manual thrust accelerates the plane forward and upward (T >> D, L > W) to gain initial altitude. During the **Glide Phase**, sustained propulsion is absent (T = 0). Gravitational potential energy (mgh) is converted into forward velocity to overcome aerodynamic drag (D):



3. Static Stability & Sink Rate Derivation

Static longitudinal stability requires the Center of Gravity (CG) to lie forward of the aerodynamic Center of Pressure (CP). The winglets provide directional yaw stability, preventing side-slip.

MINIMUM SINK RATE PRINCIPLE (TIME ALOFT OPTIMIZATION)

To maximize duration, the aircraft must minimize vertical descent velocity (v_z) rather than maximizing distance:

$$\mathbf{v_z = \sqrt{[\,(2\,m\,g) / (\rho\,S)\,]} \times [\,C_D / C_L^{3/2}\,]}$$

Maximizing the power factor ($C_L^{3/2} / C_D$) and expanding wing area (S) directly minimizes descent velocity, allowing the plane to float longer than a conventional dart.

PART C — EXPERIMENTAL DATA & ANALYSIS

1. Flight Duration Trial Table

Testing carried out indoors under wind-free conditions, releasing the aircraft from a consistent launch height and angle.

Trial #	Launch Condition	Flight Behavior & Trajectory	Duration (s)
Trial 1	Standard Manual Launch	Initial climb, transition into level glide.	[8.6]
Trial 2	Standard Manual Launch	Stable flat trajectory with low sink rate.	[8.9]
Trial 3	Standard Manual Launch	Smooth apex transition, steady descent.	[8.8]
Trial 4	Standard Manual Launch	Optimal trim glide, consistent flight path.	[9.3]
Trial 5	Standard Manual Launch	Controlled flare prior to landing.	[9.5]
Calculated Mean Flight Duration ($\bar{t}$)			**[9.1]**

2. Controlled Variable Modifications

- **Modification A (Adjusting Dihedral Angle):** Increasing upward wing angle to test for optimum roll stability versus lost vertical lift. *Control:* Keep paper mass, folding geometry, and launch velocity constant.
- **Modification B (Elevon Trim Adjustments):** Bending trailing-edge tabs upward in precise increments to optimize pitching moment and maximize $C_L^{3/2}/C_D$.

BONUS (+5 MARKS) — COMPARATIVE ANALYSIS

A secondary baseline aircraft (Classic Narrow Dart) was constructed and tested under identical laboratory conditions to evaluate performance differences.

Design Feature / Metric	Unite Plane (Glider)	Classic Dart (Baseline)	Comparison / Effect
Mean Flight Duration ($\bar{t}$)	[Duration Glider]	[Duration Dart]	Extended Hang Time
Wing Loading (W_L)	Lower (Broad Planform)	Higher (Narrow Planform)	Lower Stall Speed
Stability Characteristics	High roll/yaw stability via winglets	Fast velocity, low roll damping	Controlled Slow Glide

GENERAL SUMMARY

The structural and aerodynamic features of the Unite plane—specifically its broad wing planform, forward CG weight distribution, and integrated winglets—allow it to achieve a lower minimum sink rate (v_z) compared to standard darts, maximizing overall time aloft.